

Aditya Kulkarni

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SUMMARY

Backend and distributed systems engineer with production Python, Java, and PostgreSQL experience across fault-tolerant, high-throughput systems. Built a Java 21 + Spring Boot transactional API with PostgreSQL row-level locking and idempotency guarantees; shipped 10+ features at Isomer AI (394 commits, 7 deployments). Large-scale data processing via Kafka, Spark, and Hadoop at Capgemini. Fintech domain: Glance AI compliance RAG and PerCent HackNC24 Top 10. MS CS at NC State (May 2026).

EXPERIENCE

Product Engineer Intern	Isomer AI	May 2025 - Present
- Shipped 10+ backend features across 7 production deployments; 394 commits in TypeScript, Node.js, Express, PostgreSQL, Redis, and Docker with microservices architecture, CI/CD via GitHub Actions, and Sentry observability. - Designed REST API contracts and database schemas end-to-end; implemented RBAC, Auth0, Temporal.io workflow orchestration, and document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage).		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 - Aug 2024
Built 5+ AI-driven workflows including a health claim processing tool using OCR, NLP, and ML classification models improving accuracy by 25%; automated CI/CD on AWS and GCP, integrated Grafana monitoring, and supported model validation and red-teaming.		
Graduate Research Assistant	NC State University	Apr 2025 - Present
Designed and deployed a cloud-native genomic data platform on Google Cloud Run serving 500GB of datasets to 100+ researchers; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.		
AI/ML Intern	Capgemini	Jun 2023 - Aug 2023
Built distributed data pipelines using Hadoop and Apache Spark for large-scale batch processing and Apache Kafka streaming for real-time ingestion, reducing processing time by 15% and end-to-end latency by 20%.		

PROJECTS

Inventory Reservation Service	Java / PostgreSQL
Transactional inventory API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent POST endpoints with RFC 9457 errors and automated expiry via scheduled cleanup. Concurrency verified with Testcontainers -- two simultaneous reservations against quantity 1 produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security RBAC, Docker, GitHub Actions CI. Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Data JPA, Testcontainers, Docker	
PulseRank -- Real-Time Leaderboard	Distributed Systems
Fault-tolerant CQRS leaderboard: MySQL outbox + idempotency keys for exactly-once delivery, MongoDB TTL projections, Redis sorted sets for sub-ms reads; SSE live push; Testcontainers concurrency tests, k6 benchmarks (10K users), OpenTelemetry tracing, Prometheus + Grafana observability. TypeScript, NestJS, MySQL, MongoDB, Redis, Docker, OpenTelemetry, GitHub Actions	
Glance AI -- Fintech RAG System	Fintech / AI
Compliance-ready RAG system for fintech environments with per-tenant data isolation at the vector retrieval layer, preventing cross-tenant context leakage and LLM hallucinations; deterministic output architecture with grounded citations for compliance-sensitive financial workflows. Python, LLMs, Vector Databases, RAG, Privacy-First Architecture	
LLM Eval Harness	ML / Data Analytics
Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on 2,000 frozen examples; tracked hallucination rate, p50/p95 latency, and cost-per-call; paired bootstrap significance testing; DuckDB + Parquet columnar storage; Streamlit analytics dashboard. Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker	

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University -- BTech Computer Science CGPA: 9.32/10	Aug 2020 - May 2024
Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain	

SKILLS & AWARDS

Languages: Python, Java, TypeScript, JavaScript, SQL, C | **Backend:** Spring Boot, Node.js, Express, FastAPI, REST APIs | **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch | **Data / ML:** Kafka, Spark, Hadoop, PyTorch, Pandas | **Cloud / DevOps:** AWS, GCP, Docker, Kubernetes, CI/CD, Terraform, Grafana

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)